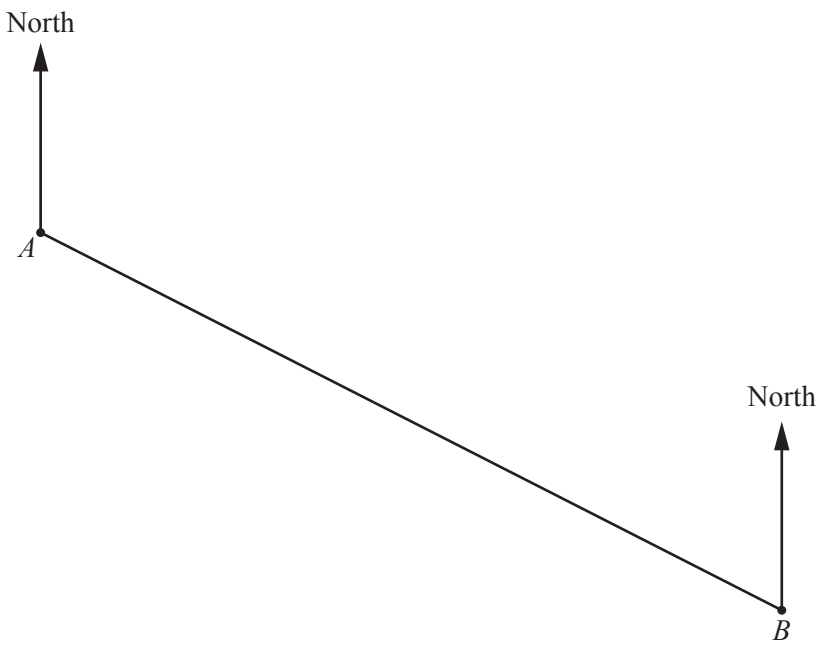
0580_s18_qp_31_Q7

7 The scale drawing shows the positions of Annika's house, *A*, and Bernhard's house, *B*, on a map.
The scale is 1 centimetre represents 300 metres.



Scale: 1 cm to 300 m

0580_s18_qp_31_Q7

(a) Work out the actual distance, in metres, between Annika's house and Bernhard's house.

..... m [2]

(b) Measure the bearing of Bernhard's house from Annika's house.

..... [1]

(c) **(i)** **Using a straight edge and compasses only**, construct the perpendicular bisector of *AB*.
Show all your construction arcs. [2]

(ii) Cordelia's house is

- the same distance from Annika's house and Bernhard's house

and

- due south of Annika's house.

Mark on the map the position of Cordelia's house.
Label this point *C*. [2]

(d) Dougie's house is

- on a bearing of 320° from Bernhard's house

and

- 1650 m from Annika's house.

Mark on the map the two possible positions of Dougie's house.
Label each of these points *D*. [4]